

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary,
Renal
Lecture: CPR 39

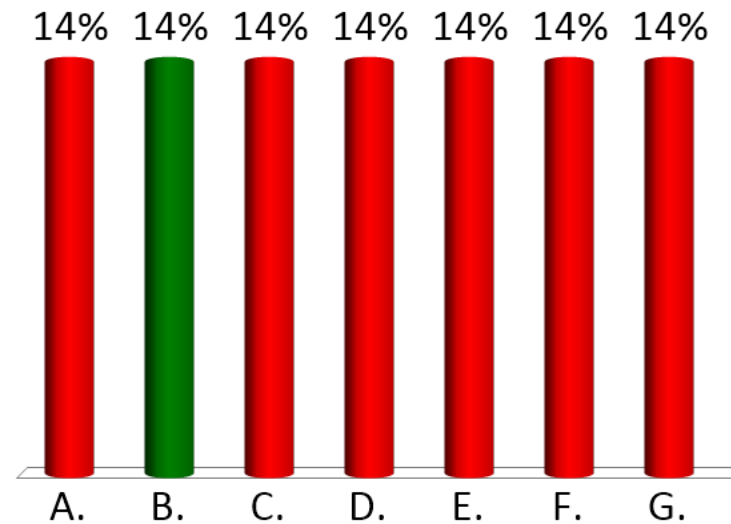
Respiratory Case Analyses

Mark Clunes, Ph.D.
St. Georges University ©



Which of the following is the most likely diagnosis in Mrs. M from the case study posted?

- A. Pulmonary embolism
- ✓ B. COPD with emphysema
- C. Asthma
- D. Pneumothorax
- E. Tension Pneumothorax
- F. Acute coronary syndrome
- G. Cardiac tamponade



The in-class session will be an analysis and MCQ session based on the following case. Please be familiar with the case and bring these slides to class. A follow up set of slides will be provided after the case discussion in the class.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0337.Discuss pathophysiologic basis of clinical features, laboratory findings and complications in a patient with chronic obstructive lung disease

Mrs M

- A 73-year-old woman comes to the physician because of a 2-day history of progressive breathlessness and difficulty breathing. She says her symptoms have worsened in the last few days when she got a cold. She has a persistent productive cough but in the last few days the sputum has been green/khaki and thick. She says that her breathlessness immediately gets worse upon sitting up and getting out of bed. She is unable to do her daily tasks without becoming very breathless and needs to sit and rest to get her breath back after walking short distances.
- She is 157 cm tall (5 ft 2 in) and weighs 40 kg (90 lbs), her BMI is 16.5 kg/m².
- She has a 7-year history of lung disease and recurrent infections with 3 hospitalizations for respiratory exacerbations in the last 18-months. Her mother died in her 70's, her sister died in her 50's and brother died in his 60's all of chronic lung disease. All had a significant smoking history. She has a 45-pack year history of cigarette smoking.
- Her temperature is 38°C (100.4°F), pulse is 98/min, blood pressure is 115/80 mm Hg and respirations are 24/min. Physical examination shows accessory muscle use with respiration, particularly on exhalation. She uses pursed lips during exhalation and has an extended expiratory time. She has central and peripheral cyanosis. Crackles are heard upon inspiration.

Mrs M continued

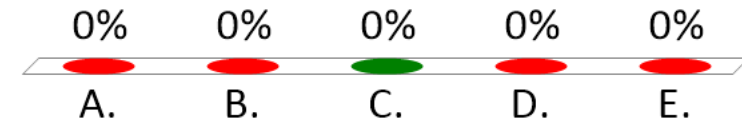
- Physical examination shows that she is alert and oriented to person, place, time and situation. Cardiac auscultation shows a normal S1 and a split S2 with a loud P2. There is jugular venous distention, and the liver is large and tender to palpation. An ECG shows mild right axis deviation, and a parasternal heave is felt. She has edema in her ankles and feet but says that the edema has only developed in the last day or so. She has reduced breath sounds and hyper-resonance to percussion in both lung fields. There is an increased antero-posterior diameter of the chest.
- Arterial blood gas analysis shows hypoxemia, hypercapnia, elevated bicarbonate, and acidemia.
- Spirometry shows she has a low FEV₁ (25% of predicted) and a low FEV₁:FVC ratio (30% of predicted). Plethysmography shows her total lung capacity is increased, and her residual volume is increased. Her vital capacity is larger than her forced vital capacity.
- An X-ray of the chest is attached.
- She is given aerosolized short acting beta agonists and systemic glucocorticoids, IV antibiotics are initiated. Supplemental oxygen is delivered via a venturi mask at a F_IO₂ of 40%.

COPD

- Obstructive – airflow limitation
- Not all COPD comes with emphysema – COPD airway disease, chronic bronchitis – emphysema lung parenchyma disease

Which of the following risk factors placed Mrs M most at risk of developing this chronic lung disease?

- A. Low BMI
- B. Family history
- ✓ C. Smoking history
- D. Heart disease
- E. Age

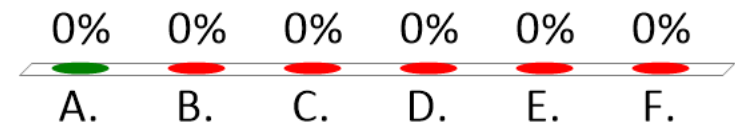


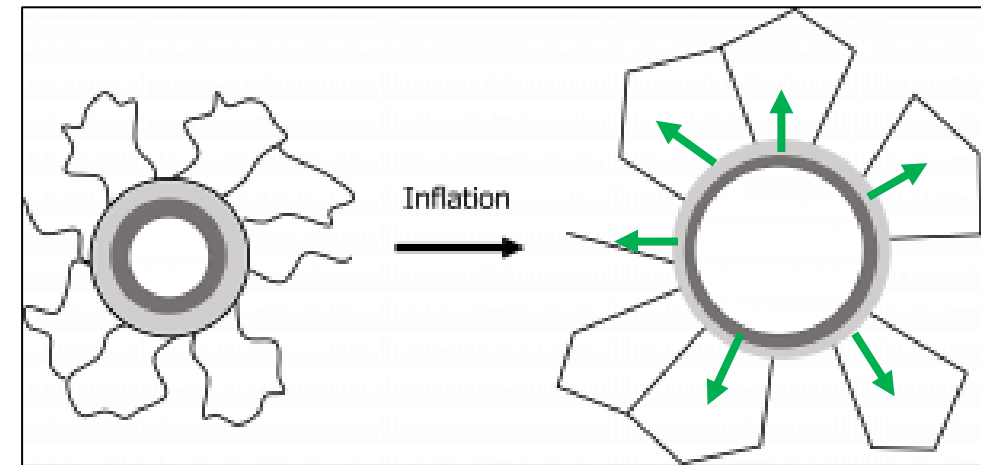
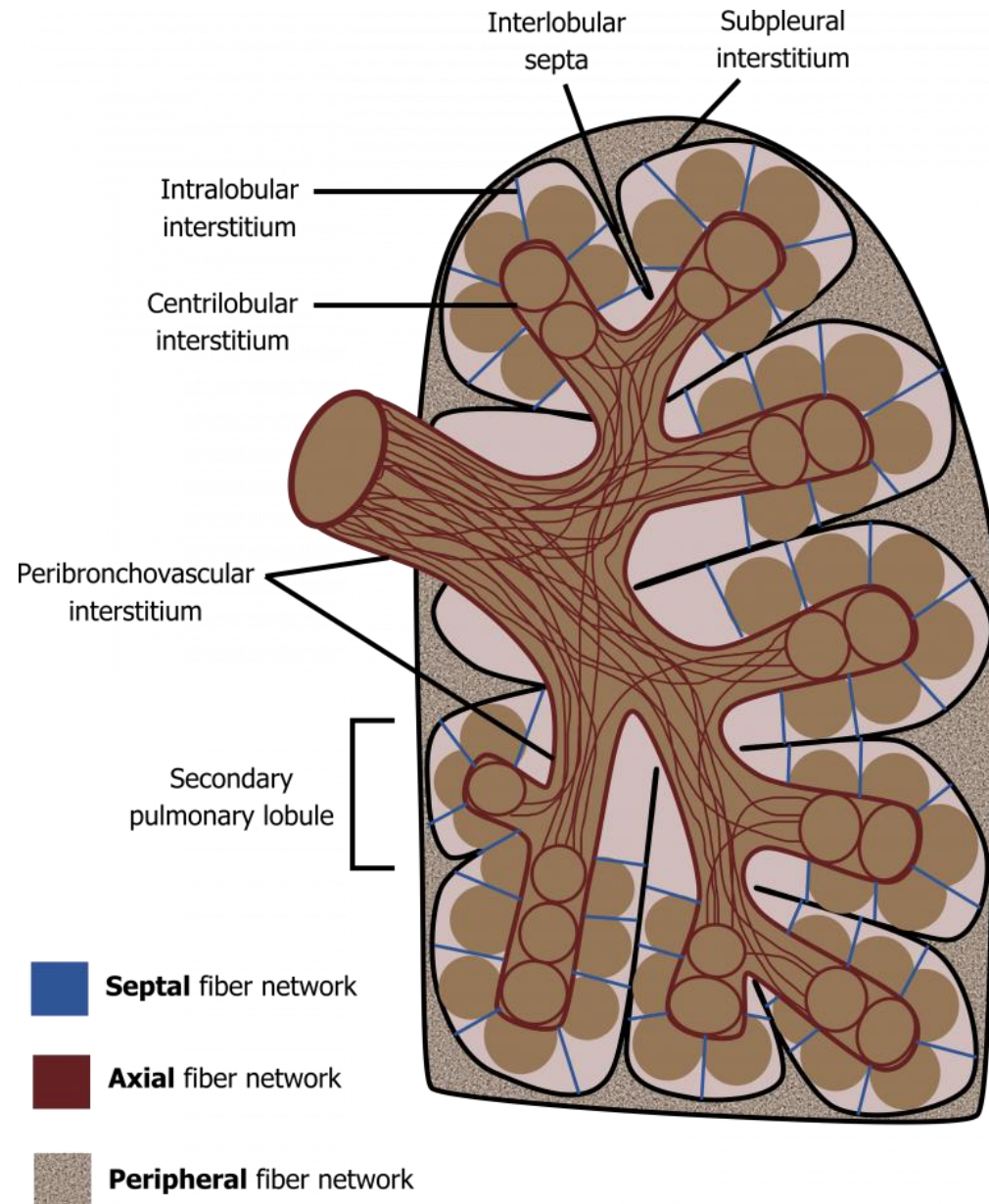
COPD - CDC

- <https://www.cdc.gov/tobacco/campaign/tips/diseases/copd.html>

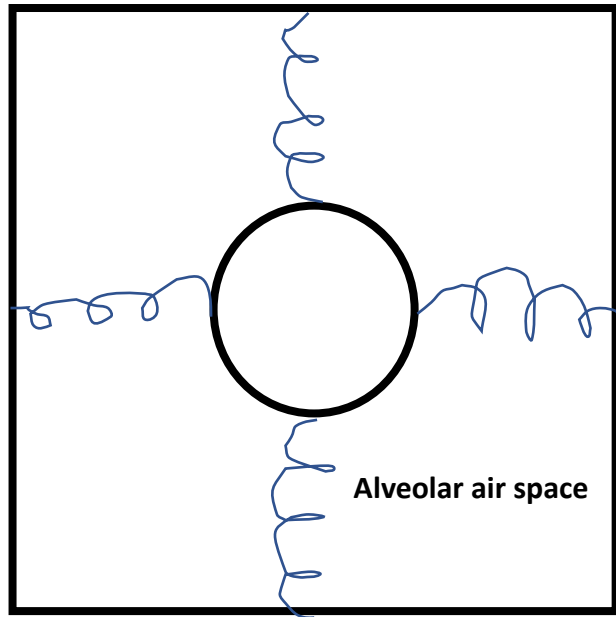
Mrs M is instructed to initiate a cough reflex after exhalation. Which of the following changes make this advice most beneficial for lung mucus clearance?

- ✓ A. Decreased radial tension
- B. Increased radial tension
- C. Increased fluid secretion
- D. Decreased fluid secretion
- E. Increased mucus secretion
- F. Decreased mucus secretion

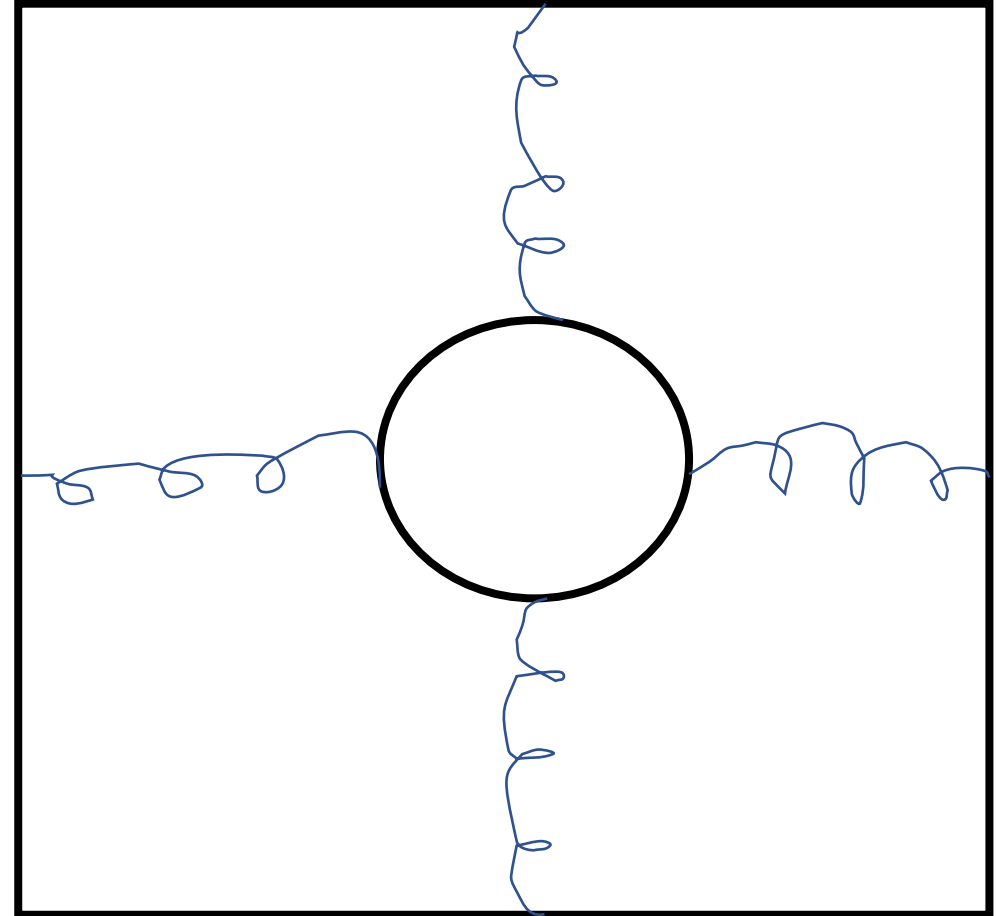




Radial Traction/Tension



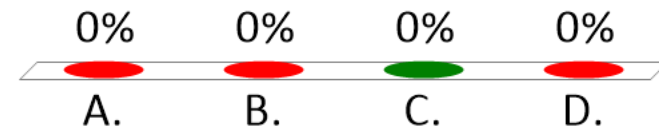
Frame with springs attached to elastic circular structure in the center. The area bounded by the springs is equivalent to the alveolar air space and the springs are equivalent to the alveolar walls.



If you expand frame (increase thorax volume) – stretches springs, they experience more tension (radial traction) and pull on the circular structure at the center. Opening it up to a larger radius

Which of the following changes occurs in expiratory airflow as a result of decreased radial tension and cough?

- A. Increased flow
- B. Decreased flow
- ✓ C. Increased velocity
- D. Decreased velocity



Flow Velocity

- Flow m^3/min
- Velocity m/min
- Area m^2

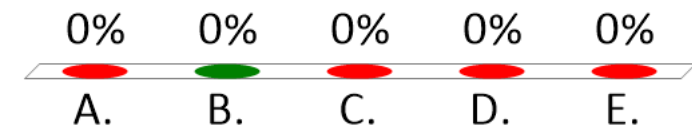
- $A * V = F$

Association between lung volume and airway caliber

- As you exhale – lung volume declines and the ‘stretch’ of the lung diminishes (radial traction).
- As the radial traction diminishes, suspended structures shrink – airways and extra-alveolar blood vessels.
- Reduced airway diameter – leads to higher airflow velocities during cough – greater chance of pulling mucus with the airflow!
- Airway milking.

Which of the following lung volumes will show the greatest fractional increase in this patient?

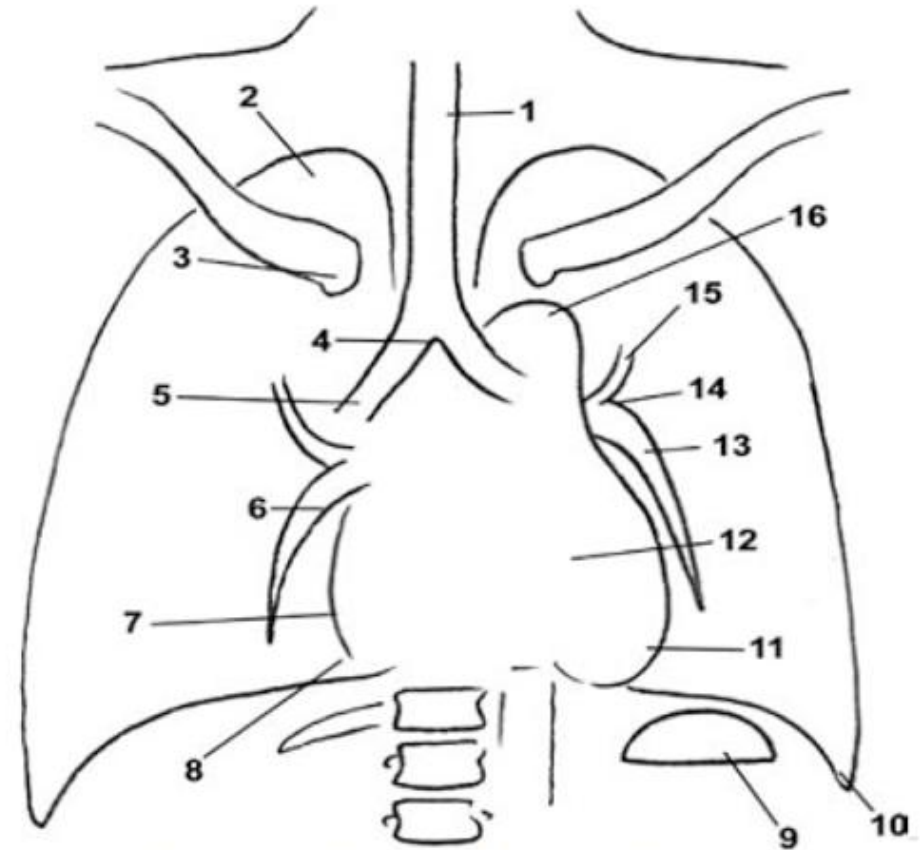
- A. Total lung capacity
- ✓ B. Residual volume
- C. Functional Residual Capacity
- D. Vital capacity
- E. Forced Vital Capacity



Chest Radiography



1. Trachea, 2. Right lung apex, 3. Clavicle, 4. Carina, 5. Right main bronchus, 6. Right lower lobe pulmonary artery, 7. Right atrium, 8. Right cardiophrenic angle, 9. Gastric air bubble, 10. Costophrenic angle, 11. Left ventricle, 12. Descending thoracic aorta, 13. Left lower lobe pulmonary artery, 14. Left hilum, 15. Left upper lobe pulmonary vein, 16. Aortic arch.



Line diagram of normal frontal chest X-ray: ^

Chest radiography



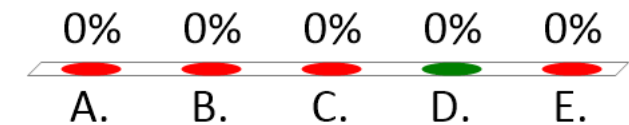
Normal



Mrs M

Which of the following radiographic features suggests airspaces in the lungs?

- A. Nodule-like structures
- B. Enlarged lung field
- C. Reduced lung fields
- ✓ D. Patchy hyper-lucency in the lung fields
- E. Patchy opacity in the lung fields



Emphysema

- Loss of lung tissue – with loss of tissue comes increased lucency.
- CT scans much better – show swiss cheese like pattern in lungs
- Smoking typically causes centrilobular emphysema.

<https://radiopaedia.org/cases/normal-chest-ct-lung-window>

<https://radiopaedia.org/cases/centrilobular-emphysema-2>

<https://radiopaedia.org/cases/normal-chest-ct-lung-window>

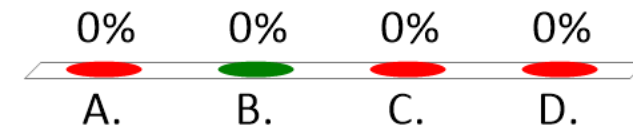
Case courtesy of A.Prof Frank Gaillard, Radiopaedia.org, rID: 8095

<https://radiopaedia.org/cases/centrilobular-emphysema-2>

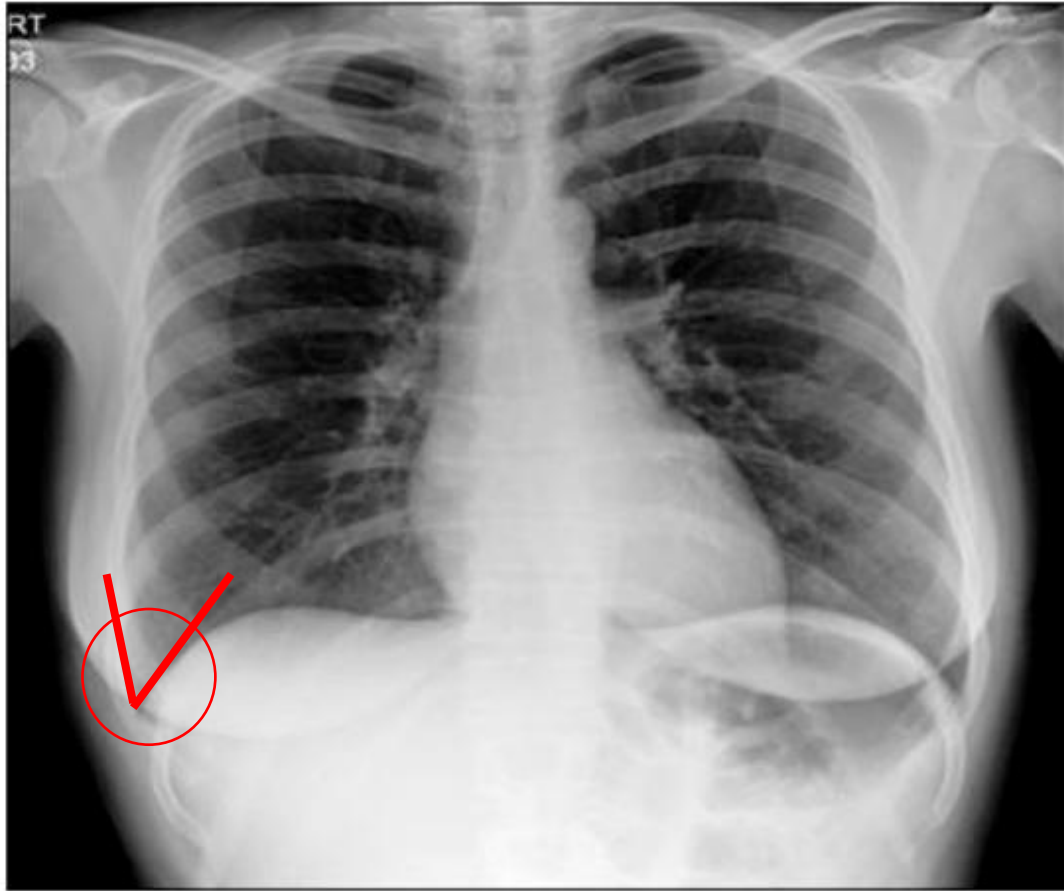
Case courtesy of Dr Alborz Jahangiri, Radiopaedia.org, rID: 46611

Which of the following radiographical findings suggests loss of lung elastance?

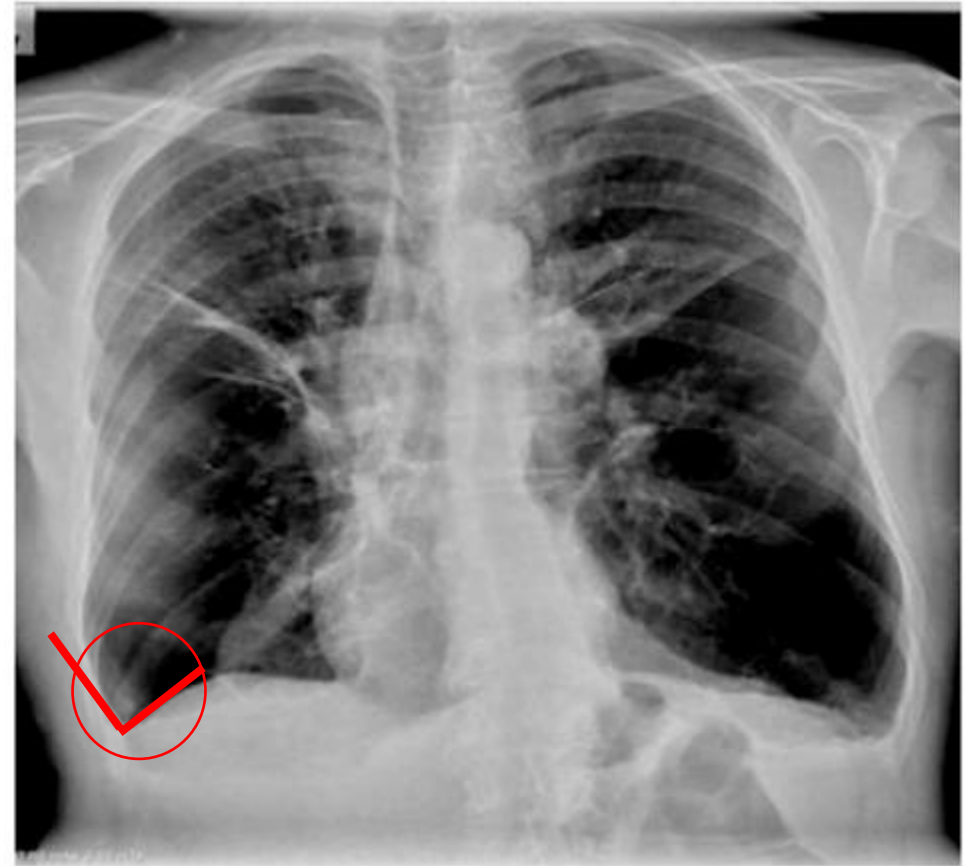
- A. Narrowing of costophrenic angle
- ✓ B. Widening of costophrenic angle
- C. Movement of the mediastinum
- D. Movement of the cardiac silhouette



Chest radiography – costophrenic angle



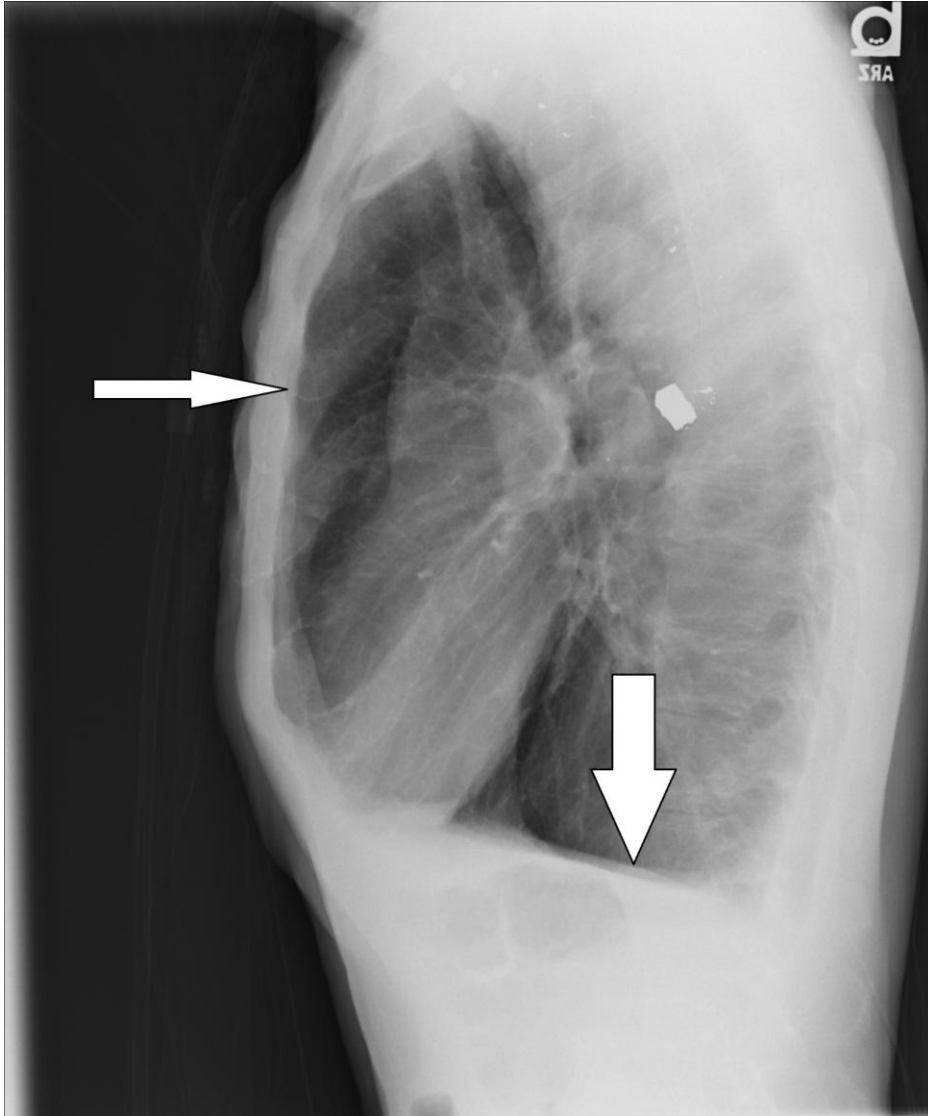
Normal



Mrs M

Loss of elastic recoil of the lungs because of emphysema allows the diaphragm to lie flatter and lower

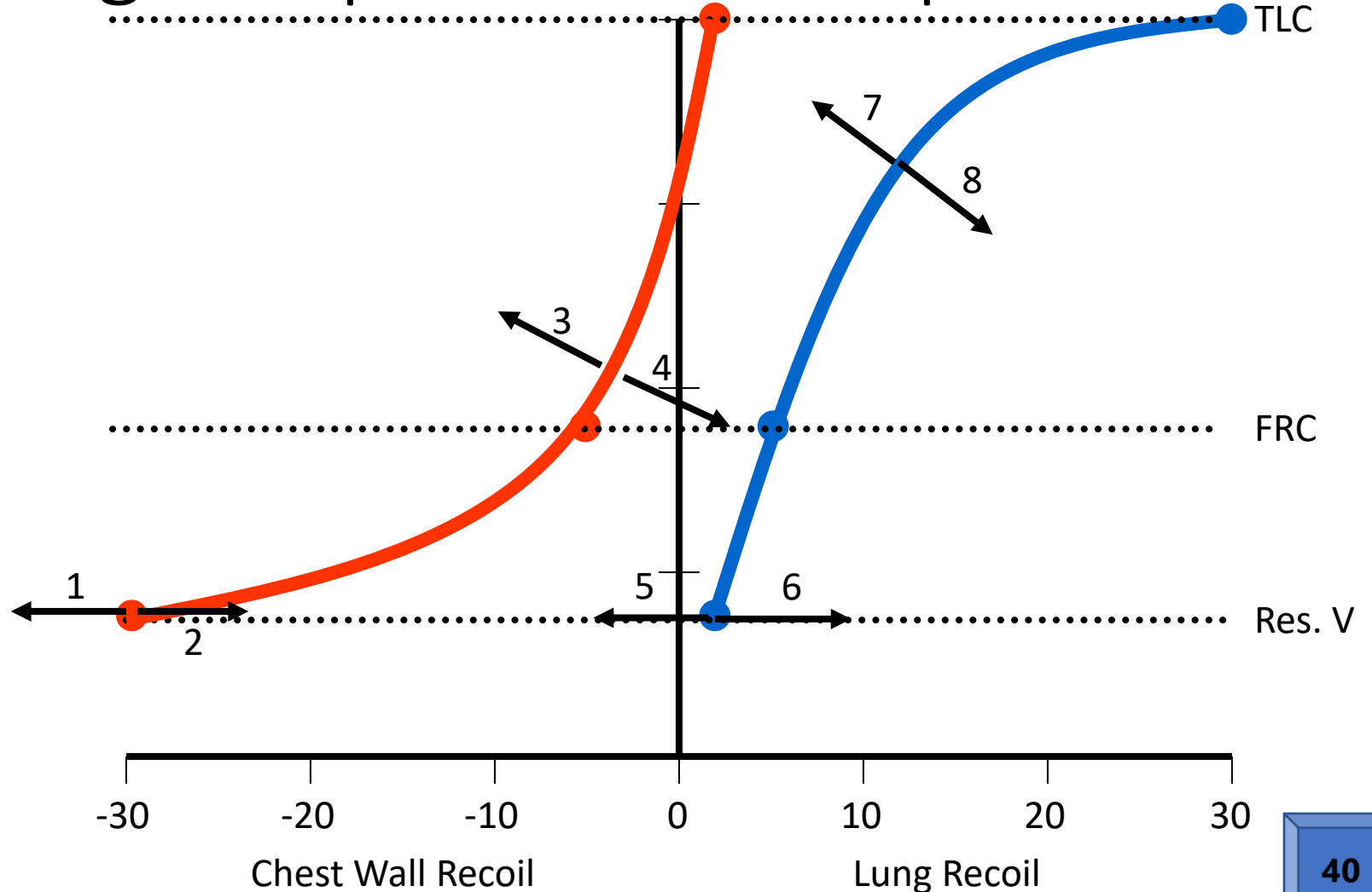
Chest radiography – lateral view



- Retrosternal translucency – sign of lung hyperextension.
- Flat diaphragm especially evident in lateral view.

Which of the following changes best explain the change in lung compliance in this patient?

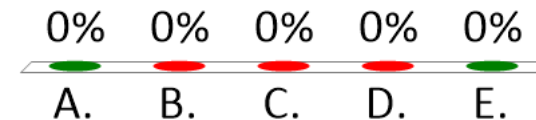
- A. 1
- B. 2
- C. 3
- D. 4
- E. 5
- F. 6
- ✓ G. 7
- H. 8



0%0%0%0%0%0%0%0%0%
A B C D E F G H

Which of the following pathophysiological mechanisms leads to increased pulmonary vascular resistance and pulmonary arterial pressures?

- ✓ A. Loss of lung tissue
- B. High RV output due to RV hypertrophy
- C. Low RV output due to RV hypertrophy
- D. Hypoxia mediated vasodilation
- ✓ E. Hypoxia mediated vasoconstriction

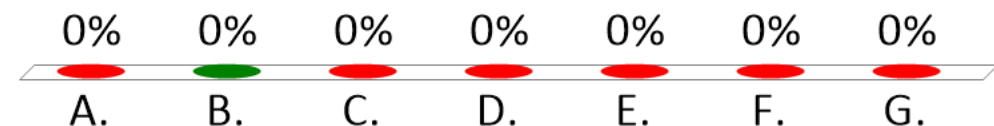


Increased pulmonary vascular resistance – causes in COPD

- Loss of lung tissue means loss of vascular tissue – and loss of vascular cross-sectional area.
- Areas of hypoxia in the lung due to poor ventilation, e.g. mucus plugs, narrow inflamed airways – lead to vascular smooth muscle contraction – hypoxic pulmonary vasoconstriction.
- Vascular smooth muscle hypertrophy due to chronic hypoxia.
- Loss of vascular tissue and hypoxia – leads to low vascular cross section and pulmonary hypertension – and increased afterload on right heart.

Which of the following physical findings suggests an increased total lung capacity?

- A. Jugular venous distention
- ✓ B. Increased anteroposterior diameter
- C. Parasternal heave
- D. Pursed lip breathing
- E. Inspiratory crackles
- F. Cyanosis
- G. Reduced breath sounds



Lung volume changes



- Chronic changes in chest architecture – ‘barrel chest’ – increased antero-posterior diameter.
- Flat low diaphragm – low lung recoil.
- Widened costophrenic angle.

Previous Exacerbation Arterial Blood Gas

	08/08	10/09	01/10	
PO ₂ (mmHg)	38	70*	72*	(75 – 100)
PCO ₂ (mmHg)	49	69	58	(35 – 45)
Arterial HCO ₃ (mmol/L)	25	32	30	(22.0 - 32.0) (22-28)
O ₂ Saturation (%)	73	94	95	(>95)
pH	7.33	7.29	7.34	(7.35 - 7.45)
*on nasal O ₂ 2L/min				
WBC - Total count (10 ⁹ /L)	12.8	12.0	12.4	(4.0 - 11.0)
Neutrophils (10 ⁹ /L)	11.3	11.1	9.1	(1.90 - 7.50)
Urea (mmol/L)	6.7	6.0	4.5	(2.5 - 7.0)
Creatinine (umol/L)	58	77	68	(40 – 100)
eGFR (mL/min/1.73m ²)	> 90	79	78	(>90)

Assume nasal cannula
gives P_IO₂ of 0.30 O₂

Does hypoventilation fully explain this patient's hypoxemia?

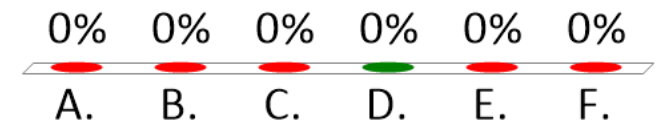
A. Yes

✓ B. No



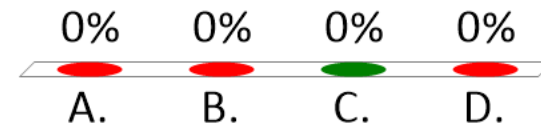
Which of the following parameters ruled out hypoventilation as the primary cause of hypoxemia in this patient?

- A. Patient has hypercapnia
- B. Patient has hypocapnia
- C. Patient has a normal A-a gradient
-  D. Patient has an increased A-a gradient
- E. Patient has acidosis
- F. Patient has alkalosis



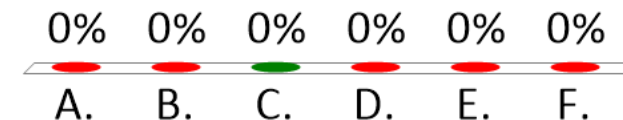
Which of the following blood gas results would most likely suggest hypoventilation as the primary cause of hypoxemia? (*note: not seen in Mrs M*)

- A. High A-a gradient and low P_aO_2 and high P_aCO_2
- B. High A-a gradient and low P_aO_2 with low P_aCO_2
- ✓ C. Normal A-a gradient with low P_aO_2 and high P_aCO_2
- D. Normal A-a gradient with high P_aO_2 and low P_aCO_2



Which of the following mechanisms best explains Mrs M's hypoxemia?

- A. Pulmonary shunt
- B. Increased diffusion distance
- ✓ C. V/Q mismatch
- D. Hypoventilation
- E. Increased dead space
- F. Pulmonary edema



She will likely have elements of all of these – but overall the V/Q mismatch is the mechanism that explains her hypoxemia

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A-a gradient

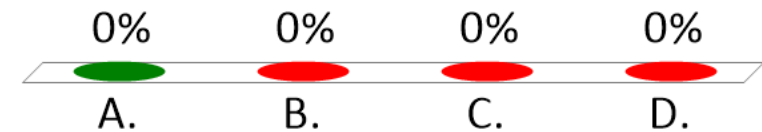
- With nasal cannula O₂ – assume P_IO₂ of 0.30 (do not need to know in this course what a fractional concentration is with different delivery!)
- $P_AO_2 = (P_B - P_{H_2O}) \times P_I O_2 - (P_aCO_2 / R)$
- $P_AO_2 = (760 - 47) \times 0.30 - (58 / 0.8)$
- $P_AO_2 = 213.9 - 72.5$
- $P_{\text{A}}O_2 = 141.4$
- $P_{\text{a}}O_2 = 72$ (10/09 visit supplemental O₂)
- **A-a** = 69.4 compared to normal 5-15 A-a

08/08 visit – no supplemental O₂
= $((760 - 47) \times 0.21) - (49 / 0.8) = 88.48$ A
With PaO₂ = 38 a
Aa gradient = 50.48

01/10 visit – supplemental O₂
= $((760 - 47) \times 0.30) - (69 / 0.80) = 127.65$ A
With PaO₂ = 70 a
Aa gradient = 57.65

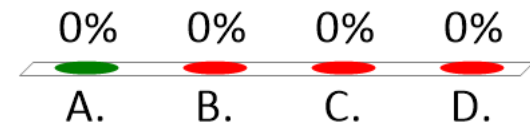
Which of the following is the primary cause of the acid base disturbance in this patient?

- A. Respiratory acidosis
- B. Respiratory alkalosis
- C. Metabolic acidosis
- D. Metabolic alkalosis



Which of the following compensations would take place for this patient's primary acid base disturbance?

- A. Increased renal reabsorption of HCO_3^-
- B. Decreased renal reabsorption of HCO_3^-
- C. Hyperventilation
- D. Hypoventilation

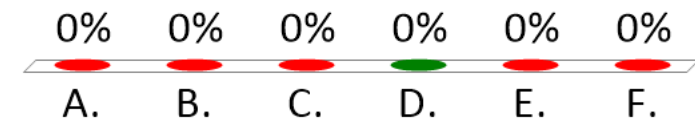


Acid base Status

- Acute respiratory acidosis - increased PCO_2 and normal bicarbonate
- Chronic respiratory acidosis – increased PCO_2 with increased renal proton secretion and increased bicarbonate reabsorption

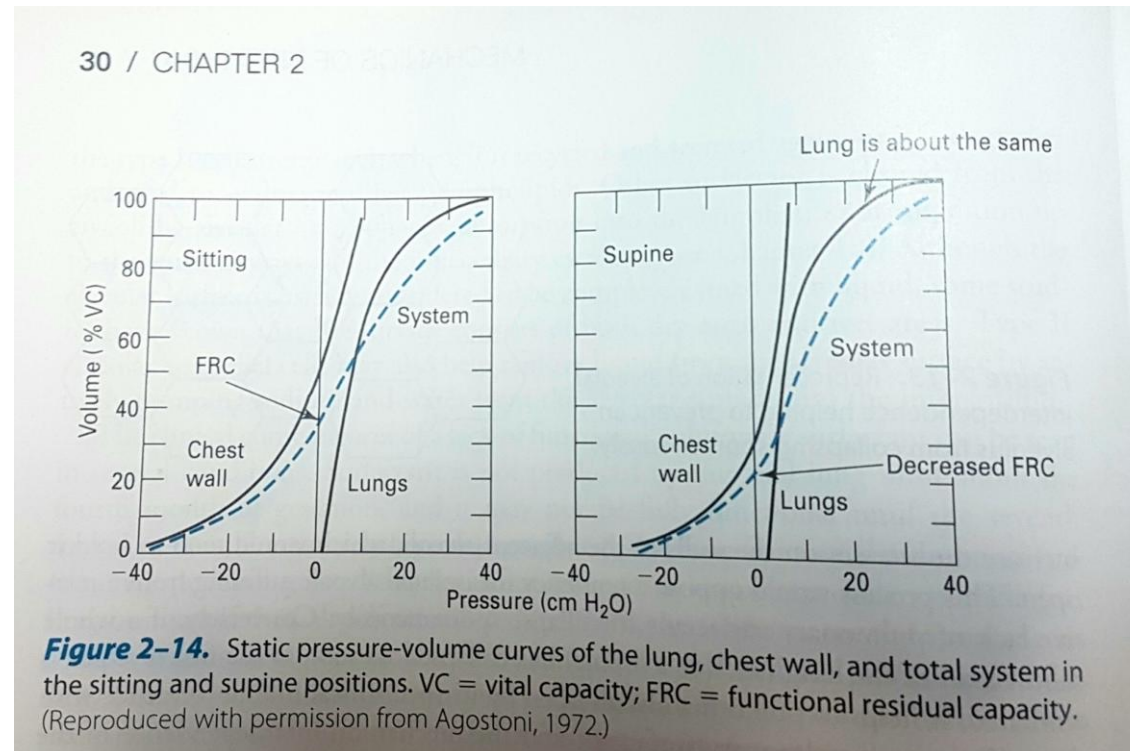
Which of the following lung volumes/capacities will increase most when Mrs M stands up?

- A. Total lung capacity
- B. Vital capacity
- C. Forced vital capacity
- ✓ D. Functional Residual Capacity
- E. Residual Volume
- F. Inspiratory Reserve Volume



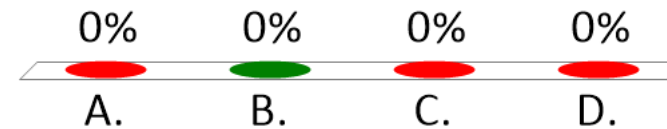
FRC

- Increases when standing over supine. Abdominal contents push into thoracic cavity when lying but are suspended and pull down on diaphragm when standing. Leads to change in shape of lung compliance curve and increase in FRC.



Which of the following factors changes most upon standing?

- A. Airflow distribution
- ✓ B. Blood flow distribution
- C. Tidal volume
- D. Minute ventilation



Blood flow is through dependent parts of lung

- Low pressure pump of RV means blood flow is highly dependent.
- Dependent – means through the lowest hanging part!
- Standing changes pulmonary blood flow patterns significantly.

The End

Pre handout follows

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Case Study

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- An X-ray of the chest is attached.
- She is given aerosolized short acting beta agonist and systemic glucocorticoids; IV antibiotics are initiated. Supplemental oxygen is delivered via a venturi mask at a $F_{I}O_2$ of 40%.



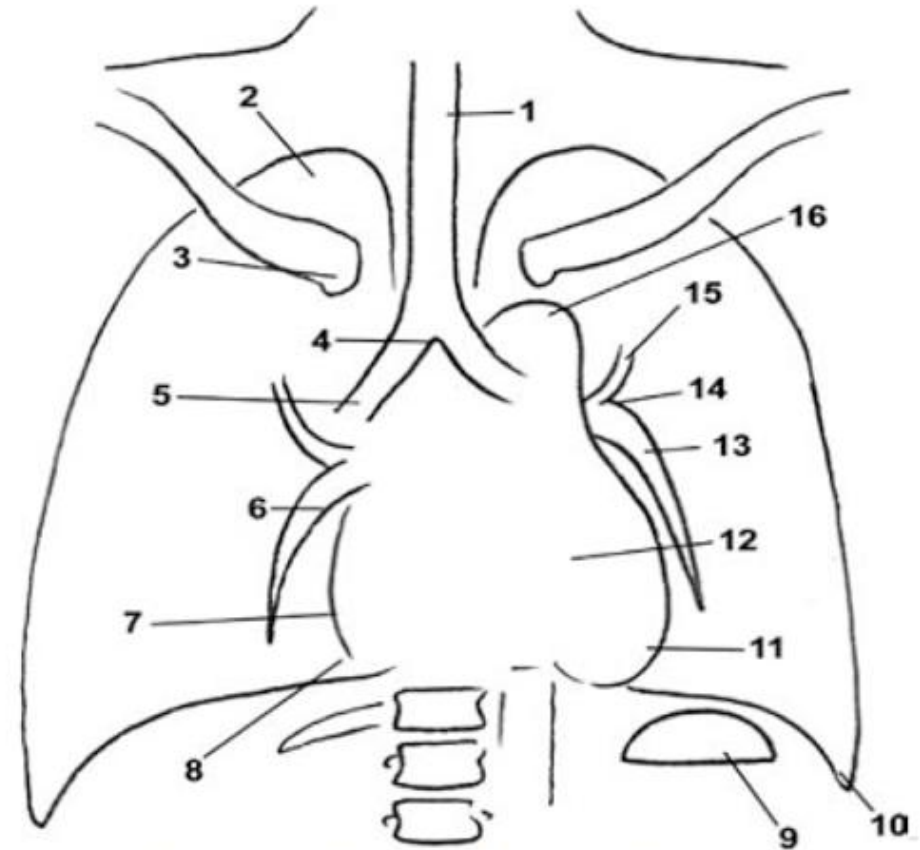
Recommended Reading

- Pulmonary Physiology Lectures, 30, 31, 34, 35, 37 and 38 and associated DLA 13, 14, 15 and 16.
- Rhoades and Bell, Medical Physiology: Principles for Clinical Medicine, Sixth Edition

Chest Radiography



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10/09 and 01/10
Assume nasal cannula
gives P_IO₂ of 0.30 O₂

Some questions to consider before the session

- What is her most likely diagnosis?
- Which features of her case back this up?
- What is her Aa gradient likely to be?
- What do you calculate her Aa gradient to be – with and without supplemental O₂.
- What kind of factors contribute to her Aa gradient?
- What is the clinical significance of this?